

Retrieval Augmented Generation (RAG)

What is RAG?

RAG enhances LLM responses by retrieving relevant documents before generating answers. It combines retrieval systems with generative AI for accurate, grounded outputs.

RAG Pipeline Steps

1. Document Loading: Ingest documents from PDFs, web, databases, files.
2. Text Splitting: Chunk documents into smaller, overlapping pieces.
3. Embedding: Convert chunks to dense vector representations.
4. Vector Store: Index vectors in a database (FAISS, Chroma, Pinecone).
5. Retrieval: Semantic search to find top-k relevant chunks for a query.
6. Generation: LLM reads retrieved context and generates the final answer.

Benefits of RAG

Reduces hallucinations by grounding answers in retrieved evidence.

Supports knowledge cutoff override - use real-time or proprietary data.

More cost-effective than fine-tuning for domain knowledge updates.

Enables source citation and auditability of responses.

RAG vs Fine-Tuning

RAG is preferred when knowledge is frequently updated or document-specific. Fine-tuning is better for adapting model style and behavior. RAG + fine-tuning can be combined for best results.